Finding anapoles from the scattering matrix

Technical note, anapole study (pysie2d), 27 October 2026

1. The question

A particle is *dark* for a given illumination when it does not scatter it: all the light that comes in leaves exactly as if the particle were not there. Such a state is called an *anapole*. We want to know, for a particle of a given shape and size, at which material permittivity ε this can happen. The method below answers this using only a standard scattering solver.

2. Describing the light around the particle

Outside the particle, any field can be written as a sum of simple wave patterns, one for each integer order $n = 0, \pm1, \pm2, \ldots$ Order n varies around the particle as $e^{in\theta}$ (θ is the angle around the particle centre), so n counts how many times the pattern oscillates around one turn. Each pattern comes in two versions:

- an **incoming** wave, travelling towards the particle, with amplitude a_n;
- an **outgoing** wave, travelling away from it, with amplitude b_n.

We keep orders $|n| \le N$ (here $N = 9$). Higher orders barely touch a particle of this size, so they can be dropped. The amplitudes then form two vectors, $\mathbf{a}$ and $\mathbf{b}$, each of length $2N + 1$.

3. The scattering matrix S

The particle is linear, so the outgoing waves depend linearly on the incoming ones:

$$\mathbf{b} = S\,\mathbf{a}.$$

S is a square matrix, here 19×19. Entry S_{mn} is how much outgoing order m is produced by a unit incoming wave of order n. If the particle were absent, every incoming wave would simply pass through and come out unchanged, so S would be the identity matrix I.

How we compute it. A solver gives the *scattered* field (total minus incident) for any illumination. We illuminate with one standing-wave pattern of order n at a time and record the scattered amplitudes of every order m. Those numbers form column n of the *scattered-wave matrix* T. Because a standing wave is half incoming and half outgoing,

$$S = I + 2T.$$

$2N + 1 = 19$ solver runs give the full matrix for one value of ε.

4. Why S is unitary

For a lossless particle (real ε), energy is conserved: the power going out equals the power coming in, for *every* incoming vector $\mathbf{a}$. In matrix terms $\|\mathbf{b}\|^2 = \|\mathbf{a}\|^2$, which holds for every $\mathbf{a}$ only if

$$S^{\mathsf{H}} S = I \qquad (S^{\mathsf{H}} \text{ is the conjugate transpose}),$$

i.e. S is *unitary*. This gives a built-in check: we compute $\max |S^{\mathsf{H}} S - I|$ and require it to be at round-off level. Measured: 2×10^{-13} for a circle, $\le 10^{-9}$ for the non-circular shapes.

5. Channels and phase shifts

A unitary matrix has a complete set of orthogonal eigenvectors $\mathbf{v}_j$, and every eigenvalue has magnitude one, so it can be written as a pure phase:

$$S\,\mathbf{v}_j = e^{2i\delta_j}\,\mathbf{v}_j.$$

- $\mathbf{v}_j$ is a **channel**: a particular mix of incoming orders that comes back out as the same mix. The particle can only delay it.
- δ_j is the channel's **phase shift**, the delay the particle imposes on it.

The scattered part of channel j is $(e^{2i\delta_j} - 1)/2$, whose size is $|\sin\delta_j|$. So:

- $\delta_j = 0$ (or a multiple of 180°): the channel passes untouched. **This is an anapole** for that channel.
- $\delta_j = \pm 90°$: the channel scatters as strongly as possible. This is a **resonance**.

6. Sweeping the permittivity

As ε increases, the particle stores more energy, and each phase shift moves steadily in one direction (it never turns back). An anapole is therefore a point where one channel's δ_j crosses zero. Between two anapoles of the same channel, its phase passes through $\pm 90°$, a resonance. So anapoles and resonances alternate.

The procedure is:

1. Compute $S(\varepsilon)$ on a fine grid (step 0.02 from $\varepsilon = 1.5$ to 25).
2. Find the eigenvalues and eigenvectors at each step.
3. Follow each channel from one step to the next by picking the eigenvector most similar to the previous one. Sorting by phase alone fails: the dropped high orders all sit at $\delta \approx 0$ and get confused with a real crossing.
4. Record where a followed δ_j changes sign. Check it by solving again at the interpolated point. A very sharp resonance can jump from $+90°$ to $-90°$ within one step and look like a sign change. At a true anapole, δ_j stays small there.

Symmetry. Our shapes are mirror-symmetric (left–right). Every channel is then either even or odd under the mirror, and the two families never mix. We split S into an even block and an odd block and treat each separately, which keeps the eigenvectors clean.

7. A given illumination

Real illumination (a plane wave, or the evanescent tail of a waveguide mode) is a fixed incoming vector $\mathbf{a}$. Writing it in channels, the scattered power is

$$P_{\text{sca}} \;\propto\; \sum_j \left|\mathbf{v}_j^{\mathsf{H}}\mathbf{a}\right|^2 \sin^2\delta_j .$$

- $|\mathbf{v}_j^{\mathsf{H}}\mathbf{a}|^2$ is how much of the illumination enters channel j.
- $\sin^2\delta_j$ is how strongly that channel scatters.

The particle is dark for $\mathbf{a}$ only if every channel that $\mathbf{a}$ feeds has a small phase shift. On a circle each order is its own channel, so one exact anapole can make one order exactly dark. A non-circular shape mixes orders, so a fixed illumination generally feeds several channels. It is then only *nearly* dark: the sum is small but not zero.

8. Checks and results

check	result
Unitarity, $\max\lvert S^{\mathsf{H}}S - I\rvert$	2×10^{-13} (circle), $\le 10^{-9}$ (shapes)
Circle anapoles vs. exact formula	all 9 recovered: 5.535, 7.053, 10.287, 14.288, 14.320, 17.869, 18.941, 23.564, 24.206
Dark design vs. direct simulation	$\times 316$ darker than the circle, both ways

Findings (silicon, $\varepsilon = 12.08$, equivalent radius 500 nm, $\lambda = 1550$ nm):

- Neither optimised dark shape has an anapole at $\varepsilon = 12.08$. An interior calculation puts the nearest at 11.95 and 14.15. Off the circle, these are points where a channel's δ_j comes close to zero (0.05, 0.015) without reaching it. Following channels through near-crossings is fragile: on these shapes it produced false crossings, so non-circular roots must be confirmed by an interior calculation.
- The design that is dark above the waveguide is only $1.7\times$ darker than a circle when the same evanescent wave hits it in free space, against $27\times$ above the slab. Light reflected by the slab is part of the cancellation.

Limits. The method needs a lossless particle (otherwise S is not unitary and the phases are not pure). It describes the particle alone in free space. Results are from one solver (pysie2d) and are not externally validated.